

SkySense-VITA: Towards Universal In-context Segmentation of Multi-modal Remote Sensing Imagery

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<https://kang-wu.github.io/SkySense-VITA>

Abstract

While recent foundation models for remote sensing segmentation have shown notable progress, they still fall short in processing diverse multi-modal inputs, synergizing complementary prompt types, and leveraging semantic hierarchies. To address these limitations, we introduce SkySense-VITA, a unified in-context segmentation model, which synergistically processes both optical and Synthetic Aperture Radar (SAR) imagery using *Visual*, *Textual*, or fused prompts. Based on a novel prompt-and-prediction decoupling strategy, we propose the VITA-Former and VITA-Decoder to decouple multi-modal prompt fusion and prediction process, allowing the model to flexibly support visual-only, textual-only, and fused prompt modes. We train SkySense-VITA with a progressive two-stage strategy: a first stage of Image-Level Alignment Pretraining featuring optical-SAR alignment, and a second stage of Pixel-Level In-context Pretraining using Semantic Granularity Annealing (SGA), a coarse-to-fine curriculum that enables robust hierarchical learning. To support this training, we introduce our new large-scale, multi-modal Sky-VT-300k dataset. Extensive experiments show SkySense-VITA establishes a new state-of-the-art (SOTA) on 18 datasets, with an average performance lead of over 10% mean Intersection over Union (mIoU).

1. Introduction

Semantic segmentation of remote sensing imagery is a fundamental technology for Earth observation, critical for applications like urban planning and disaster response [2, 47, 52]. However, the dominant “one model, one task” paradigm relies on specialized models trained with expensive, large-scale annotated datasets. This dependency results in poor generalization to new geographic regions, sensor types, or classes, hindering scalable planetary analysis.

Recently, researchers have begun developing remote

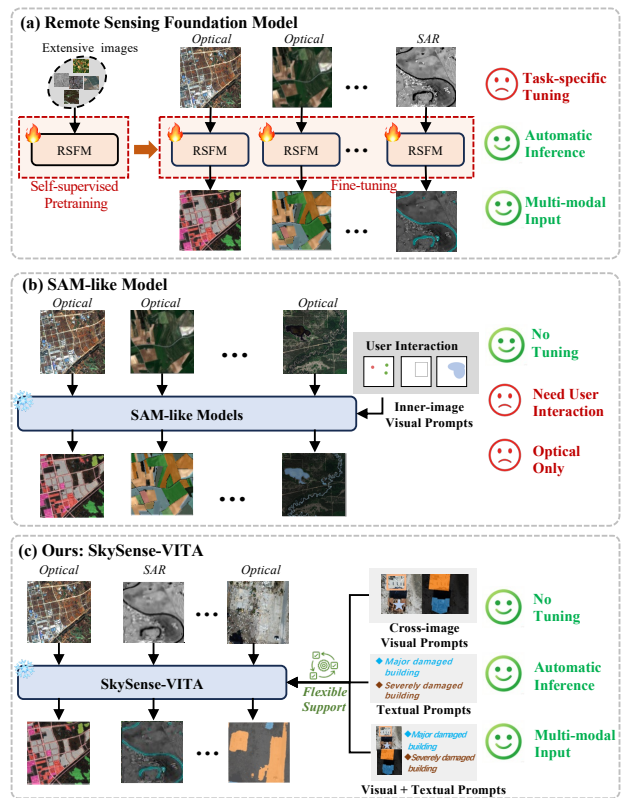


Figure 1. Comparison of remote sensing image segmentation paradigms. (a) Conventional remote sensing foundation models require task-specific fine-tuning, limiting their flexibility. (b) SAM-like models require user interaction at inference time, cannot leverage textual prompts, and lack native support for SAR imagery. (c) Our proposed SkySense-VITA is tuning-free, operates without per-image user interaction, and supports both optical and SAR remote sensing imagery.

sensing foundation models (RSFM) [6, 9, 12, 18, 21, 24, 26, 31, 39, 50], typically by pretraining on massive unlabeled datasets to learn transferable representations (Fig. 1(a)). Although these RSFMs can be extended to multi-modal inputs and boost downstream performance [53], they still suffer

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from a critical limitation: their adaptation requires a time-consuming and labor-intensive fine-tuning process, which is impractical for time-critical applications. More recently, a new frontier has emerged with fine-tuning-free universal segmentation models. A representative work is the Segment Anything Model (SAM) [14, 35] shown in Fig. 1(b), which achieves rapid task adaptation via per-image user interaction without parameter modification. However, SAM is limited for remote sensing, as it is trained on natural images and struggles with the domain gap, especially for SAR images. Furthermore, its reliance on per-image user interaction prevents large-scale automated processing. While pioneering works [17, 52, 56, 60] have explored prompt-based architectures for universal segmentation in remote sensing, they suffer from three key limitations:

First, limited modality support. Remote sensing imagery is inherently multi-modal, with optical sensors providing rich textural details and SAR offering all-weather, all-time capability [9]. A unified segmentation model should support both modalities for robust deployment, while current methods [17, 60] are designed exclusively for optical imagery, rendering them inapplicable to SAR imagery and limiting real-world application.

Second, limited prompt type support. Existing universal models in remote sensing are divided into visual-prompt-based [52] and textual-prompt-based methods [17, 56, 60], each with distinct limitations. Visual prompts, which use reference images and masks, provide rich visual context but suffer from task granularity ambiguity. For instance, a *car* annotation could also mean the *vehicle*, or the *transportation* super-category. Textual-prompt-based methods suffer from semantic ambiguity. For instance, a single term like *plant* can refer to either *vegetation* or a *power plant*. They also fail to capture vast intra-class visual variance, as text alone cannot encode specific stylistic nuances, such as the dramatically different appearance of *houses* across continents [43]. Consequently, jointly supporting both prompt types is crucial for flexible and robust generalization.

Third, limited semantic hierarchy consideration. Remote sensing categories are naturally hierarchical (e.g., *car* $\subset$ *vehicle*). Existing methods [17, 52, 56] typically treat classes as a flat list, ignoring crucial structural knowledge and forcing the model to learn difficult, fine-grained classes from scratch without the benefit of coarse-level contextual priors. It also exacerbates inter-class confusion, thereby compromising the structure and robustness of the feature space.

To address these limitations, we introduce SkySense-VITA, a unified in-context segmentation model that processes both optical and SAR imagery and supports visual, textual, and fused prompts, where “in-context” means segmentation conditioned on external prompts (visual exemplars, textual descriptors, or their combinations) without task-specific fine-tuning. To train model at scale, we con-

struct Sky-VT-300k, a large dataset of optical and SAR remote sensing images with pixel-level semantic annotations and aligned visual exemplars and textual descriptors, covering more than 300,000 samples across 176 categories. Building on SkySense’s decoupled backbone [9], we introduce learnable modality tokens that aggregate modality-specific features across layers, improving robustness under heterogeneous sensors. We further develop VITA-Former and VITA-Decoder based on a decoupling strategy for prompts and predictions: VITA-Former performs prompt decoupling, fusing visual and textual cues while retaining modality-specific pathways; VITA-Decoder performs prediction decoupling, jointly optimizing the visual, textual, and fused branches to mitigate cross-branch interference and maximize per-branch performance. Finally, we introduce a progressive two-stage pretraining pipeline: 1) Image-Level Alignment Pretraining, which aligns optical and SAR representations via cross-modal learning; and 2) Pixel-Level In-Context Pretraining with Semantic Granularity Annealing (SGA), a semantic-tree-guided coarse-to-fine curriculum that reduces semantic conflicts and improves generalization. Experiments on in-distribution and out-of-distribution benchmarks show that SkySense-VITA consistently outperforms prior models across datasets and metrics. Our main contributions are:

- We introduce SkySense-VITA, a unified in-context segmentation model that jointly supports optical/SAR inputs and visual, textual, and fused prompting.
- We develop a progressive pretraining strategy named SGA to obtain robust, hierarchy-aware representations.
- Extensive evaluations on 18 datasets demonstrate SOTA performance, achieving an average gain exceeding 10% mIoU compared to existing baselines.

2. Related Work

2.1. Remote Sensing Foundation Model

RSFMs excel at learning robust feature representations from extensive unlabeled data via self-supervised learning. Predominant methods include contrastive learning [9, 12, 24, 26, 50] and Masked Image Modeling [6, 18, 21, 31, 39], with some works combining both [27, 29]. Despite excellent generalization on diverse data, these RSFMs necessitate task-specific fine-tuning for downstream segmentation tasks. To overcome it, we propose SkySense-VITA, an in-context segmentation model that handles diverse remote sensing tasks without fine-tuning.

2.2. In-context Segmentation

In-context learning was popularized by GPT-3 [1] and adapted to vision by Painter [48]. SegGPT [49] further modeled in-context segmentation via in-context coloring, enabling few-shot generalization without fine-tuning.

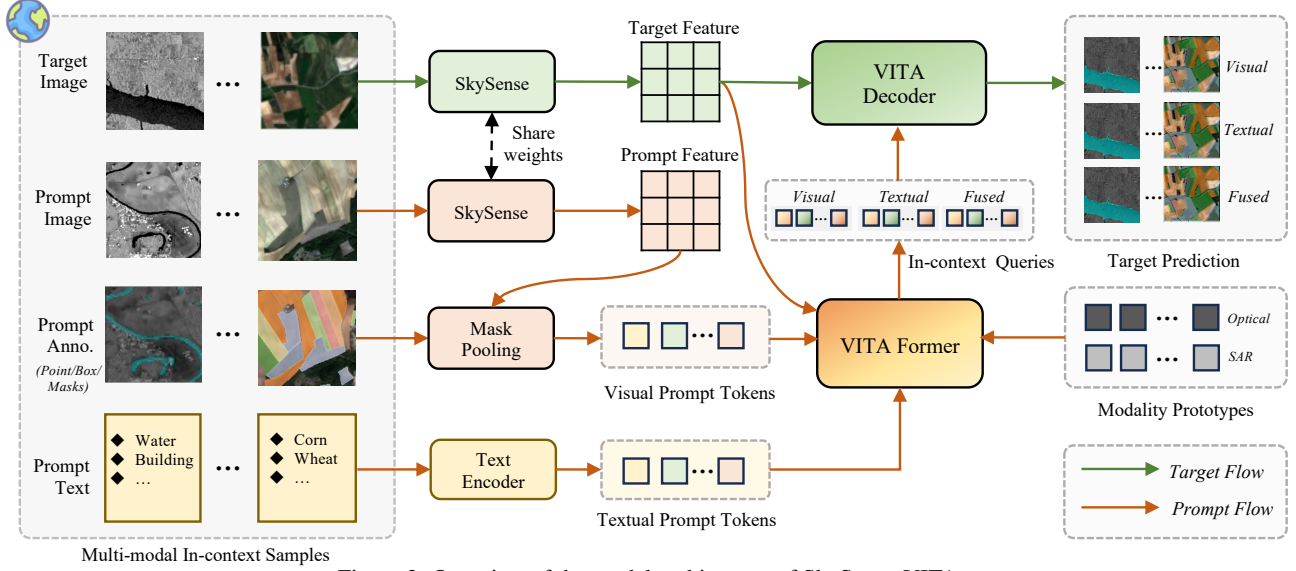


Figure 2. Overview of the model architecture of SkySense-VITA.

Subsequent work falls into three paradigms: (i) SAM-based [23, 38, 40, 57], which generate prompts via cross-image correspondence via DINOv2 [32] and segment with SAM [14] but incur high compute; (ii) query-based [15, 22], which use MaskFormer-style decoupling and prompt engineering to handle diverse tasks; and (iii) response-based [28], which design specialized decoders to refine cross-image correspondences. In remote sensing, SkySense++ [52] introduces semantic-band to transfer in-context learning to randomized index segmentation, improving segmentation performance.

2.3. Open-Vocabulary Segmentation for Remote Sensing

Open-vocabulary semantic segmentation aims to segment arbitrary categories in the open world by textual prompts. In remote sensing, SegEarth-OV [17] is the first training-free open-vocabulary semantic segmentation framework, introducing a feature upsampler to recover spatial detail. SkySense-O [60] further adopts a vision-centric modeling principle, leveraging knowledge graph to correct alignment ambiguities induced by language guidance.

Most above methods support only one prompt type and therefore inherit its limitations. Although some [13, 36] combine visual and textual prompts, they operate on per-image prompts in natural images and lack mechanisms to synergize both. In contrast, SkySense-VITA tackles the underexplored cross-image in-context setting, supporting visual, textual, and fused prompts across optical and SAR via prompt-prediction decoupling for robust fusion.

3. SkySense-VITA

In this section, we present the pretraining dataset, model architecture and pretraining details of SkySense-VITA.

3.1. Pretraining Dataset

For Image-level Alignment Pretraining, we create a dataset of co-located optical-SAR-text triplets by adding SAR imagery with the optical imagery from [60]. For pixel-level in-context pretraining, we construct Sky-VT-300k dataset, a diverse dataset of over 300,000 samples aggregated from 12 sources [8, 20, 30, 41, 44–46, 51, 52, 54, 60]. This dataset spans 7 satellite platforms and 176 classes, covers both optical and SAR modalities, and provides pixel-level masks with aligned visual exemplars and textual descriptors for in-context training.

3.2. Model Architecture

In Fig. 2, our architecture has two flows: a target flow encodes the target image and generates segmentation masks, and a prompt flow encodes visual and/or textual prompts to produce in-context queries. The overall pipeline consists of three parts: Multi-modal Feature Extraction, Multi-modal Prompt Fusion, and Prompt-Guided Decoding.

Multi-modal Feature Extraction. Given a target image x_{tgt} and a prompt pair $(x_{pmt}, v_{pmt}, t_{pmt})$, where x_{pmt} is the prompt image, v_{pmt} is its visual annotation, and t_{pmt} is the class name. We extract features F_{tgt} and F_{pmt} using a factorized encoder [9] from optical and SAR inputs. We construct visual prompt tokens $P_V \in \mathbb{R}^{N \times C}$ by class-wise masked average pooling from F_{pmt} and v_{pmt} . Textual prompt tokens $P_T \in \mathbb{R}^{N \times C}$ are obtained by encoding category names t_{pmt} with a pretrained text encoder [33].

Multi-modal Prompt Fusion by VITA-Former. To flexibly support visual-only, textual-only, and fused prompting, VITA-Former performs early fusion over visual, textual, and fused prompt tokens (Fig. 3). We first mitigate the optical/SAR discrepancy by injecting modality prototypes P_M into the visual tokens via cross-attention, yield-

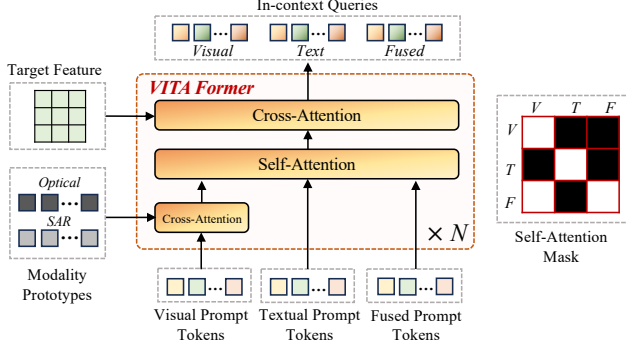


Figure 3. Overview of VITA-Former for feature fusion.

ing domain-adapted P'_V . Here P_M are learnable modality-aware tokens (two prototypes for optical and SAR) that are shared across layers and updated end to end. We then concatenate $[P'_V, P_T, P_F]$, where the fused tokens P_F are initialized from P_T by a linear projection. A masked self-attention enforces the following: (i) P'_V and P_T are isolated to preserve uni-modal specificity, and (ii) P_F can attend to both to absorb cross-modal cues. Finally, cross-attention with target features F_{tgt} produces three in-context prompt queries $\{Q_{ctx}^k\}_{k \in \{V, T, F\}}$ corresponding to the visual (V), textual (T), and fused (F) modes.

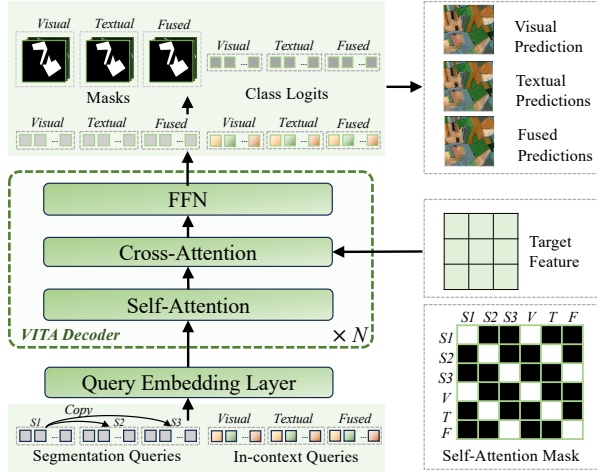


Figure 4. Overview of VITA-Decoder for decoupled decoding.

Prompt-Guided Decoupled Decoding by VITA-Decoder. VITA-Decoder in Fig. 4 follows a query-based design [3, 4] with explicit mode decoupling. We replicate learnable segmentation queries into three copies, $Q_{seg}^V, Q_{seg}^T, Q_{seg}^F$, and pair them with the in-context queries Q_{ctx}^k . All queries $Q = \{Q_{seg}^k, Q_{ctx}^k\}_{k \in \{V, T, F\}}$ are first concatenated, passed through a Query Embedding Layer for modality and type encoding, and then refined by a decoder stack $\mathcal{D}$ using: (i) block diagonal masked self-attention to isolate modes and reduce cross-branch interference, and (ii) cross-attention to F_{tgt} . Let the refined queries be $Q_{seg}^{k'} and $Q_{ctx}^{k'}$. Mask predictions M_k are obtained follows Mask2Former [4]:$

$$M_k = \mathcal{F}_{mask}(Q_{seg}^{k'}, \phi(F_{tgt})), \quad k \in \{V, T, F\}, \quad (1)$$

where $\phi(\cdot)$ projects encoder features to the mask space. $\mathcal{F}_{mask}$ maps query tokens to mask embeddings and computes per-pixel logits via a dot product with $\phi(F_{tgt})$. For class assignment, we replace a fixed classifier with fully prompt-guided classification using similarity to $Q_{ctx}^{k'}$:

$$\mathcal{P}_k = \text{softmax}(\text{norm}(Q_{seg}^{k'}) \text{norm}(Q_{ctx}^{k'})^\top / \tau), \quad (2)$$

where $\text{norm}(\cdot)$ denotes ℓ_2 normalization and τ is a temperature for scaling the logits. The final output $\tilde{y}_k$ is obtained by combining M_k and $\mathcal{P}_k$.

3.3. Pretraining

Our pretraining employs a progressive two-stage pipeline. The first stage focuses on Image-level Alignment Pretraining to bridge the domain gap between the optical and SAR. The second stage, Pixel-level In-context Pretraining, learns to segment prompt-specified categories under a semantic hierarchy, enabling coarse-to-fine learning with visual and textual prompts.

Image-level Alignment Pretraining. To efficiently align SAR to the pretrained optical-text space, we adopt an indirect alignment strategy [25]. The optical encoder E_{opt} is frozen and only the SAR encoder E_{SAR} is trained. Given a co-registered pair (x_{opt}, x_{SAR}) , we create two augmented SAR views x_{SAR}^a and x_{SAR}^b and extract global embeddings $\{f_{opt}, f_{SAR}^a, f_{SAR}^b\}$ via global average pooling after feature extraction. We optimize two objectives: an intra-modal contrastive loss $\mathcal{L}_{CL}$ on (f_{SAR}^a, f_{SAR}^b) to learn robust SAR features, and a cross-modal alignment loss $\mathcal{L}_{CMA}$ between f_{SAR}^a and the fixed f_{opt} to anchor SAR to the optical space [16]. The stage loss $\mathcal{L}_{img}$ is:

$$\mathcal{L}_{img} = \lambda_{CL} \mathcal{L}_{CL}(f_{SAR}^a, f_{SAR}^b) + \lambda_{CMA} \mathcal{L}_{CMA}(f_{SAR}^a, f_{opt}), \quad (3)$$

where λ_{CL} and λ_{CMA} balance the terms.

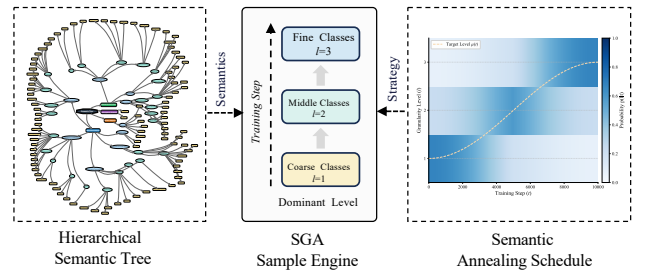


Figure 5. Overview of Semantic Granularity Annealing.

Pixel-level In-Context Pretraining. Remote sensing categories are hierarchical. Inspired by [37, 60], we use Qwen-3 [55] to construct a semantic tree $\mathcal{T}$ over C classes with depth L , where leaves are fine-grained categories and ancestors are coarser super-classes shown in Fig. 5. For any leaf c_i , let its ancestral path be $\mathcal{A}(c_i) = \{c_i^{(1)}, \dots, c_i^{(L)}\}$ from coarse to fine, with $c_i^{(L)} = c_i$. We propose Semantic Granularity Annealing (SGA) to achieve coarse to fine semantic

Methods	Publication	Optical										SAR
		FLAIR	Potsdam	FBP	LoveDA	iSAID	JL-16	OAM-TCD	GF1-WHU	AIR-PV	QuickQuake Buildings	ETCI-Flood
<i>In-context Segmentation Model (One-shot)</i>												
Painter [48]	CVPR23	22.48	11.10	6.31	8.58	22.33	5.33	20.52	3.70	3.10	30.18	-
SegGPT [49]	ICCV23	25.99	27.82	19.87	31.44	30.11	10.17	35.80	27.39	38.42	7.04	-
PerSAM [57]	ICLR24	23.91	24.02	11.82	27.31	25.34	9.09	28.21	18.36	37.32	27.31	-
DINOv [15]	CVPR24	34.39	47.07	18.28	25.50	25.94	9.15	37.05	27.61	26.46	30.56	-
SINE [22]	NIPS24	30.23	23.94	14.67	19.89	20.96	10.21	28.78	23.21	18.80	23.91	-
SEGIC [28]	ECCV24	39.17	58.80	24.02	34.02	46.41	10.41	41.36	31.91	32.12	35.48	-
SkySense++* [52]	NMI25	44.50	62.70	41.62	47.25	47.74	20.13	43.39	20.73	47.51	46.94	-
<i>Open-vocabulary Segmentation Model (Zero-shot)</i>												
SegEarth-OV [17]	CVPR25	32.19	34.80	20.02	32.01	19.93	9.80	4.31	23.44	5.25	28.12	-
SkySense-O [60]	CVPR25	27.01	23.42	8.09	34.61	20.93	1.30	33.14	18.24	35.52	28.55	-
SkySense-VITA(Visual)	-	53.29	71.34	47.04	56.11	56.07	23.81	45.69	70.12	56.55	75.85	27.40
SkySense-VITA(Text)	-	63.02	81.53	58.87	60.54	61.55	36.66	47.23	69.43	57.87	75.59	29.27
SkySense-VITA(Visual+Text)	-	64.03	81.94	57.26	62.08	59.89	35.97	46.78	71.27	57.55	75.25	29.58

Table 1. Comparison of segmentation performance across different in-distribution datasets. Values represent mIoU (%). * indicates that this model was retrained on the fully aligned training dataset. - indicates that the model does not support this dataset.

learning. Unlike hard-staged approaches, SGA introduces a smooth, probabilistic curriculum to prevent catastrophic forgetting for coarse classes of some datasets. We first define a continuous target granularity $\mu(t) \in [1, L]$:

$$\begin{aligned}\gamma(t) &= \frac{1 - \cos(\pi \cdot \min(1, t/T))}{2}, \\ \mu(t) &= 1 + \gamma(t)(L - 1),\end{aligned}\quad (4)$$

where T is the total training steps. $\mu(t)$ transitions from 1 (coarse) at $t = 0$ to L (fine) at $t = T$. At step t , we sample a discrete level $l \in \{1, \dots, L\}$ from a probability distribution $p(l | t)$, which is centered at $\mu(t)$ and controlled by a temperature τ_{SGA} :

$$p(l | t) = \frac{\exp(-|l - \mu(t)|/\tau_{\text{SGA}})}{\sum_{j=1}^L \exp(-|j - \mu(t)|/\tau_{\text{SGA}})}.\quad (5)$$

A pixel x with leaf label c_i is then trained to predict its ancestor at the sampled level, $y = c_i^{(l)}$.

As visualized in Fig. 5 for $L=3$, $T=10,000$, and $\tau_{\text{SGA}} = 0.95$, early training favors coarse categories and gradually shifts to fine categories. This reduces forgetting and stabilizes training, especially for mixed datasets with heterogeneous granularity depths.

The pixel-level objective follows set prediction [4] and sums over the three output modes $k \in \{V, T, F\}$. For each mode, we perform Hungarian matching against the level- l targets by SGA and compute the pretraining loss $\mathcal{L}_{\text{pixel}}$:

$$\begin{aligned}\mathcal{L}_{\text{pixel}} &= \sum_{k \in \{V, T, F\}} \sum_{i=1}^{N_q} [\lambda_{\text{cls}} \mathcal{L}_{\text{cls}}(\mathcal{P}_k(i), c_{\hat{\sigma}(i)}^{(l)}) \\ &\quad + \lambda_{\text{mask}} \mathcal{L}_{\text{mask}}(M_k(i), a_{\hat{\sigma}(i)}^{(l)})],\end{aligned}\quad (6)$$

where N_q is the number of query, $\mathcal{L}_{\text{cls}}$ is cross-entropy over level- l classes and $\mathcal{L}_{\text{mask}}$ combines Dice and Focal losses. Unmatched queries are supervised as no-object as in [4].

4. Experiments

4.1. Implementation Details

Pretraining. The first Image-level Alignment Pretraining runs on a cluster of 80 A100 GPUs. The second Pixel-Level In-context Pretraining runs on 8 A100 GPUs. For this stage, we train the model on our new Sky-VT-300k dataset for 10,000 iterations with a batch size of 64. More details, including optimizer settings and learning rate schedules, are included in the supplementary materials.

Architecture. We employ the factorized SkySense encoder from [9] and the pretrained text encoder from CLIP [33]. The total model size is 1.1B parameters. This is composed of 969M parameters for the image encoder and 149M for the text encoder. The VITA-Former and VITA-Decoder are lightweight and efficient, comprising only 18M parameters.

4.2. Performance on In-distribution Tasks

We adopt FLAIR [8], Potsdam [10], FBP [44], LoveDA [46], iSAID [51], JL-16 [52], OAM-TCD [45], GF1-WHU [20], AIR-PV [54], QuickQuake Buildings [41], and ETCI-Flood [30] datasets for in-distribution evaluation. The pretraining datasets and evaluation benchmarks are completely disjoint to avoid data leakage. We report mIoU of in-context segmentation models under one-shot setting and open-vocabulary segmentation models under zero-shot setting. For our SkySense-VITA, we apply these one-shot and zero-shot settings to its visual-only and textual-only modes, respectively.

Tab. 1 show that natural-image in-context models (e.g., Painter, SegGPT) transfer poorly to remote sensing tasks. Furthermore, while remote sensing open-vocabulary seg-

Methods	Publication	Domain generalization				Class generalization		
		Vaihingen	DeepGlobe	GLH-Water	C2S-SAR	FloodNet	SOS	GLVM-Post
<i>In-context Segmentation Model (One-shot)</i>								
Painter [48]	CVPR23	5.96	19.51	35.96	-	24.84	-	6.82
SegGPT [49]	ICCV23	47.84	23.04	61.93	-	38.90	-	39.45
PerSAM [15]	CVPR24	18.18	31.24	52.66	-	34.39	-	3.30
DINOv [57]	ICLR24	33.29	42.95	64.42	-	34.69	-	3.94
SINE [22]	NIPS24	23.57	33.46	50.18	-	33.10	-	2.68
SEGIC [28]	ECCV24	59.45	37.16	51.54	-	60.02	-	5.18
SkySense++* [52]	NMI25	55.37	49.63	74.28	-	54.92	-	18.50
<i>Open-vocabulary Segmentation Model (Zero-shot)</i>								
SegEarth-OV [17]	CVPR25	31.66	52.58	12.91	-	43.13	-	3.32
SkySense-O [60]	CVPR25	28.24	17.14	24.40	-	29.87	-	4.91
SkySense-VITA(Visual)	-	51.62	47.02	82.02	21.18	53.19	34.49	14.20
SkySense-VITA(Text)	-	60.94	51.27	86.09	16.83	57.52	14.19	44.97
SkySense-VITA(Fused)	-	62.75	53.06	86.85	16.82	62.19	39.25	46.08

Table 2. Comparison of segmentation performance across out-of-distribution datasets, including domain and class generalization evaluation. Values represent mIoU (%). * indicates that this model was retrained on the fully aligned training dataset. - indicates that the model does not support this dataset.

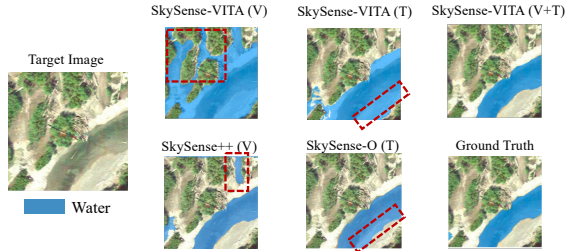


Figure 6. Qualitative out-of-distribution results on GLH-Water. Our model consistently produces more precise and complete masks for both unseen domains.

mentation models like SkySense-O [60] are trained on remote sensing data, they rely solely on text prompts and negative sampling, without visual references, leading to weak per-class in-context performance.

For a fair comparison, we retrain SkySense++ on Sky-VT-300k with aligned settings. The results demonstrate that SkySense-VITA constantly outperforms SkySense++ in visual-only, textual-only, and fused-prompt modes, achieving an average improvement of over 18%.

In terms of SAR support, competing methods struggle with dual-polarization(VV/VH) inputs from ETCI-Flood [30]. SkySense-VITA, benefiting from its natively supports SAR imagery, establishes a strong new benchmark, achieving 29.58 % mIoU in fused-prompt mode. Due to low-quality visual prompts for small-sized categories in 1-shot settings [42], fusion occasionally underperforms textual-only mode, which can be mitigated by increasing visual prompt number (see Supplementary Materials).

4.3. Performance on Out-of-distribution Tasks

We evaluate out-of-distribution generalization from two critical perspectives. First, we assess domain generalization on [5, 7, 11, 19], where test datasets do not overlap with the training data, but their categories are seen during training. For example, the Vaihingen [11] shares classes with the

Potsdam [10] but differs significantly in domain (e.g., sensor characteristics, geographic location). This setup stringently evaluates a model’s ability to generalize to new domains for known categories. Second, we assess class generalization on [34, 58, 59]. In these tasks, both the domain and the majority of classes are novel (e.g., *flooded road* is not present in the training set). This evaluation measures the model’s generalization capability across categories.

Table 2 presents the results for both domain and class generalization. In the domain generalization evaluation, generalist models trained on natural images continue to exhibit limited performance. The in-context segmentation methods, benefiting from the inclusion of domain-specific visual information, achieve better results. For instance, SkySense++ leads the best-performing open vocabulary model, SkySense-O, by 49.88% on GLH-Water dataset, highlighting the advantage of in-context visual prompts for domain adaptation. Our SkySense-VITA demonstrates competitive performance under the visual-prompt setting, and under textual and fused prompts consistently outperforms all other methods, demonstrating superior domain generalization ability. In the class generalization tasks, SkySense-VITA demonstrates an outstanding capability for discovering new classes, which we attribute to its synergistic support for both visual and textual prompts. This is particularly evident on the FloodNet dataset, where the fused prompt mode achieves an mIoU of 62.19 %, surpassing the best single-modality prompt, textual-only, by over 4%.

We provide a qualitative comparison by visualizing the outputs of various models on the GLH-Water datasets in Fig. 6. The visual results confirm our quantitative findings: our model produces significantly cleaner and more accurate segmentation masks for both unseen domains and novel classes, whereas competing methods often suffer from segmentation artifacts or missed detections.

4.4. Discussions & Ablation Studies

Effectiveness of VITA-Former. We validate the VITA-Former’s effectiveness in Tab. 3. The baseline (*w/o* VITA-Former) feeds prompt tokens to the decoder and uses a “naive” fusion (channel-concat&conv) for the fused mode. Our full model with VITA-Former is superior across all modes (visual, textual, and fused). Crucially, the baseline’s naive fusion often performs worse than its best single-modality input. Our VITA-Former avoids this by selectively integrating features to resolve conflicts. To more intuitively illustrate this, we provide a visualization of the response maps in Fig. 7, showing how the prompt tokens from the VITA-Former attend to the image features. We show examples for *bridge* on iSAID and *tree* on FloodNet. The results show that after passing through the VITA-Former, the prompt tokens for the textual and fused branches take more attention to the target region, demonstrates the VITA-Former’s effectiveness in fusing and refining prompt information.

Dataset	Method	V-only	T-only	V+T
Potsdam	w/o VITA-Former	68.72	80.27	79.49
	w/ VITA-Former	71.34	81.53	81.94
ETCI-Flood	w/o VITA-Former	27.63	14.15	20.26
	w/ VITA-Former	27.40	29.27	29.58
FloodNet	w/o VITA-Former	49.48	55.19	47.96
	w/ VITA-Former	53.19	57.52	62.19

Table 3. Ablation study on the VITA-Former: Visual-only (V), Textual-only (T), and Fused (V+T). All metrics are mIoU (%).

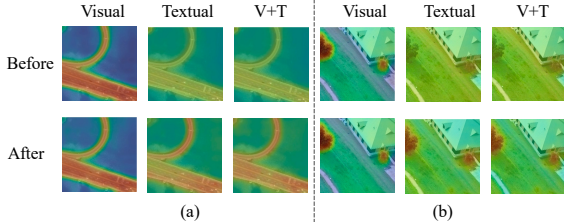


Figure 7. Response maps showing how prompt tokens attend to image features. After the VITA-Former, the textual (T) and fused (V+T) prompt tokens attend more strongly to their target regions: (a) *bridge* (iSAID) and (b) *tree* (FloodNet).

Impact of Decoupled Decoding Our prompt-guided decoupled decoding is the key to co-optimizing the visual-only, textual-only, and fused output modes within a single model. This design allows for the joint optimization of all prompt combinations while maintaining robust performance even when only a single modality is available at inference. To validate this, we conduct an ablation study comparing our full model with a baseline trained only on fused visual and textual prompts, which bypasses our decoupled design. The results in Tab. 4 show that the baseline model suffers a severe performance collapse when presented with only visual or textual prompts at inference time, with its mIoU dropping by over 24%. In contrast, our

Method	V-only	T-only	Fused
w/o Decoupled	33.12	34.57	59.31
w/ Decoupled	53.19	57.52	62.19

Table 4. Ablation study on the Prompt-Guided Decoupled Decoding strategy on the FloodNet dataset. The baseline is trained only on fused prompts. All metrics are mIoU (%).

model with decoupled decoding achieves strong results in both uni-modal settings, notably retaining 57.52% mIoU on FloodNet with only the textual prompt. Furthermore, our approach also yields a performance gain in the fused-prompt mode. This clearly demonstrates the outstanding flexibility and effectiveness of our decoupled decoding design, which successfully optimizes for all three modes simultaneously.

Method	ETCI-Flood	FloodNet
w/o Prototypes	25.81	59.12
w/ Prototypes (Add)	14.39	50.45
w/ Prototypes (Cross-Attn)	29.58	62.19

Table 5. Ablation study on modality prototypes (mIoU (%)).

Ablation Study on the Modality Prototypes. We verify the effect of introducing modality prototypes (P_M) in the VITA-Former. Tab. 5 shows the results. Adding the prototypes brings a 3.77% mIoU improvement to the SAR dataset (ETCI-Flood) and 3.07% to the optical dataset (FloodNet), demonstrating the effectiveness of explicitly injecting modality-specific information. We also compare our cross-attention fusion method with a simple “add” operation. The results show that directly adding the prototypes degrades performance. This suggests that the prototypes are not just learning a simple offset but are encoding rich, modality-specific information that must be fused intelligently via attention, rather than naively combined.

Effectiveness of Image-level Alignment Pretraining. We assess the Image-level Alignment Pretraining for the SAR encoder by comparing an aligned model with an unaligned baseline that directly inherits the SAR encoder from SkySense (no Stage 1). All other settings are matched. Table 6 reports mIoU on ETCI-Flood and C2S-SAR. Aligned implementation yields consistent gains: +2.76 mIoU on ETCI-Flood and +3.14 mIoU on C2S-SAR, indicating that image-level alignment narrows the optical–SAR gap.

Method	ETCI-Flood	C2S-SAR
Unaligned	26.82	13.68
Aligned	29.58	16.82

Table 6. Effectiveness of Image-Level Pretraining for the SAR encoder. All metrics are mIoU (%).

Ablation Study on the SGA Strategy. We assess SGA on iSAID and FloodNet. Unless noted, SGA uses a cosine schedule with depth $L = 3$, temperature $\tau = 0.95$, and total steps $T = 10000$ as defined in Sec. 3.3. Table 7 shows that

SGA improves performance by +2.18 mIoU on iSAID and +1.16 mIoU on FloodNet, supporting the effectiveness of coarse-to-fine hierarchical training.

Method	iSAID	FloodNet
w/o SGA	57.71	61.03
w/ SGA	59.89	62.19

Table 7. Ablation study on the Semantic Granularity Annealing (SGA) strategy. All metrics are mIoU (%).

Impact of Point/Box/Mask. Our mask pooling extraction method is flexible, allowing us to accept points, boxes, or masks as visual prompts. We test this capability on iSAID and FloodNet. For “Point”, we randomly sample 5 points from the ground truth mask and use the same points across all compared methods for fairness. For “Box”, we use the tight axis-aligned (minimum enclosing horizontal) bounding box. Tab. 8 shows that the performance of our model consistently improves as the visual prompt becomes more precise (from Point to Box to Mask). Furthermore, fusing the textual prompt provides an additional boost across all three visual prompt types, again confirming the benefit of our multi-modal fusion.

Prompt Type	iSAID		FloodNet	
	V-only	V+T	V-only	V+T
Point (5)	23.97	45.83	29.16	54.40
Box	24.81	49.23	29.34	55.41
Mask	56.07	59.89	53.19	62.19

Table 8. Performance with visual-only (V) and fused (V+T) prompts. Values represent mIoU (%)

Comparison with SAM Series. SAM [14, 35] is designed for intra-image visual prompting, whereas SkySense-VITA is optimized for inter-image “in-context” segmentation. To compare, we adapt our model to an intra-image setting. As shown in Tab. 9, SAM outperforms SkySense-VITA in the visual-only mode, which is expected given SAM’s high optimization for this specific task. However, SkySense-VITA’s strengths are evident in its flexibility: unlike SAM, it natively supports SAR and can fuse textual prompts. The results show that when textual prompts are integrated, SkySense-VITA significantly improves over 27% mIoU in all setting and even surpasses the performance of SAM2 by average 7.52 in mIoU, demonstrating the power of our multi-modal fusion. The qualitative results in Fig. 8 confirm that even in 1-shot setting, our model achieves excellent accuracy, and can be further refined in difficult scenes by adding more visual points.

Prompt Scaling Performance A key feature of visual prompting is the ability to scale performance by increasing number of prompts. We investigate this scaling behavior on FloodNet dataset. Fig. 9 shows that for both visual-only and fused modes, performance steadily increases as more visual prompt examples are added, eventually plateauing after 8

Method	Point(1)	Point(5)	Point(10)	Point(50)
SAM	52.33	65.42	67.89	67.95
SAM2	46.52	68.22	71.21	66.60
SkySense-VITA(V)	27.84	32.09	43.27	50.51
SkySense-VITA(V+T)	56.84	72.99	75.25	77.57

Table 9. Comparison with SAM and SAM2 on intra-image prompting. Values represent mIoU (%).

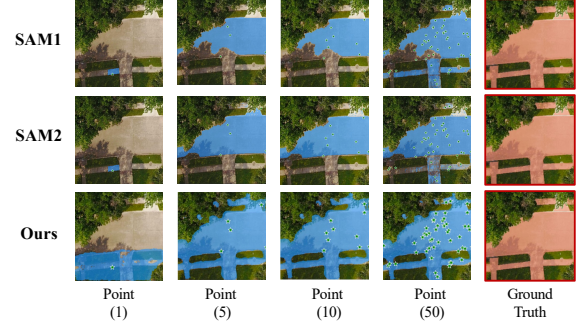


Figure 8. Qualitative comparison with SAM/SAM2. Both point and textual prompts are used for SkySense-VITA.

prompts. The fused mode is consistently superior to the visual-only mode. However, its growth curve is less steep, as the textual prompt already provides a strong semantic prior, and the additional visual prompts serve to refine an already-strong baseline.

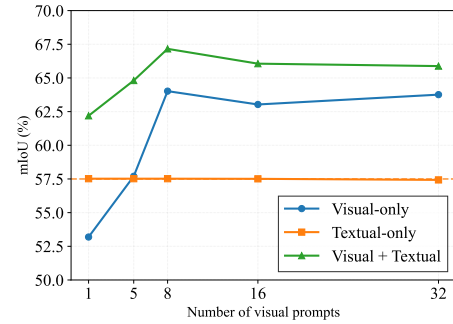


Figure 9. Performance scaling as the number of visual prompts (masks) increases, for both visual-only and fused modes.

5. Conclusion

We introduced SkySense-VITA, a unified in-context segmentation model to synergistically handle optical and SAR imagery using visual, textual, or fused prompts. Our model addresses key prior limitations through two main contributions: (1) a novel prompt-and-prediction decoupling architecture (VITA-Former and VITA-Decoder) for robust, interference-free co-optimization, and (2) a progressive training strategy featuring SGA for robust coarse-to-fine learning. Supported by our new Sky-VT-300k dataset, SkySense-VITA establishes a new SOTA on 18 segmentation tasks of remote sensing imagery. For future work, we plan to extend our model to diverse Earth observation sensors and temporal dynamics.

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References

- [1] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020. 2
- [2] Wei Chen, Lorenzo Bruzzone, Bo Dang, Yuan Gao, Youming Deng, Jin-Gang Yu, Liangqi Yuan, and Yansheng Li. Rest: Holistic learning for end-to-end semantic segmentation of whole-scene remote sensing imagery. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. 1
- [3] Bowen Cheng, Alex Schwing, and Alexander Kirillov. Per-pixel classification is not all you need for semantic segmentation. *Advances in neural information processing systems*, 34:17864–17875, 2021. 4
- [4] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1290–1299, 2022. 4, 5
- [5] Cloud to Street, Microsoft, and Radiant Earth Foundation. A global flood events and cloud cover dataset (version 1.0). Radiant MLHub, 2022. <https://registry.opendata.aws/c2smsfloods/>. 6
- [6] Yezhen Cong, Samar Khanna, Chenlin Meng, Patrick Liu, Erik Rozi, Yutong He, Marshall Burke, David Lobell, and Stefano Ermon. Satmae: Pre-training transformers for temporal and multi-spectral satellite imagery. *Advances in Neural Information Processing Systems*, 35:197–211, 2022. 1, 2
- [7] Ilke Demir, Krzysztof Koperski, David Lindenbaum, Guan Pang, Jing Huang, Saikat Basu, Forest Hughes, Devis Tuia, and Ramesh Raskar. Deepglobe 2018: A challenge to parse the earth through satellite images. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 172–181, 2018. 6
- [8] Anatol Garioud, Nicolas Gonthier, Loic Landrieu, Apolline De Wit, Marion Valette, Marc Poupée, Sébastien Giordano, et al. Flair: a country-scale land cover semantic segmentation dataset from multi-source optical imagery. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., 2024. 3, 5
- [9] Xin Guo, Jiangwei Lao, Bo Dang, Yingying Zhang, Lei Yu, Lixiang Ru, Liheng Zhong, Ziyuan Huang, Kang Wu, Dingxiang Hu, et al. Skysense: A multi-modal remote sensing foundation model towards universal interpretation for earth observation imagery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27672–27683, 2024. 1, 2, 3, 5
- [10] ISPRS. 2d semantic labeling contest - potsdam. <https://www.isprs.org/resources/datasets/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx>, 2018. 5, 6
- [11] ISPRS. 2d semantic labeling contest - vaihingen. <https://www.isprs.org/resources/datasets/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx>, 2018. 6
- [12] Pallavi Jain, Bianca Schoen-Phelan, and Robert Ross. Self-supervised learning for invariant representations from multi-spectral and sar images. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 15:7797–7808, 2022. 1, 2
- [13] Qing Jiang, Feng Li, Zhaoyang Zeng, Tianhe Ren, Shilong Liu, and Lei Zhang. T-rx2: Towards generic object detection via text-visual prompt synergy. In *European Conference on Computer Vision*, pages 38–57. Springer, 2024. 3
- [14] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4015–4026, 2023. 2, 3, 8
- [15] Feng Li, Qing Jiang, Hao Zhang, Tianhe Ren, Shilong Liu, Xueyan Zou, Huaizhe Xu, Hongyang Li, Jianwei Yang, Chunyuan Li, et al. Visual in-context prompting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12861–12871, 2024. 3, 5, 6
- [16] Junnan Li, Ramprasaath Selvaraju, Akhilesh Gotmare, Shafiq Joty, Caiming Xiong, and Steven Chu Hong Hoi. Align before fuse: Vision and language representation learning with momentum distillation. *Advances in neural information processing systems*, 34:9694–9705, 2021. 4
- [17] Kaiyu Li, Ruixun Liu, Xiangyong Cao, Xueru Bai, Feng Zhou, Deyu Meng, and Zhi Wang. Segearth-ov: Towards training-free open-vocabulary segmentation for remote sensing images. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 10545–10556, 2025. 2, 3, 5, 6
- [18] Xuyang Li, Danfeng Hong, and Jocelyn Chanussot. S2mae: A spatial-spectral pretraining foundation model for spectral remote sensing data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24088–24097, 2024. 1, 2
- [19] Yansheng Li, Bo Dang, Wanchun Li, and Yongjun Zhang. Gilh-water: A large-scale dataset for global surface water detection in large-size very-high-resolution satellite imagery. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 22213–22221, 2024. 6
- [20] Zhiwei Li, Huanfeng Shen, Huifang Li, Guisong Xia, Paolo Gamba, and Liangpei Zhang. Multi-feature combined cloud and cloud shadow detection in gaofen-1 wide field of view

- imagery. *Remote sensing of environment*, 191:342–358, 2017. 3, 5
- [21] Zhihao Li, Biao Hou, Siteng Ma, Zitong Wu, Xianpeng Guo, Bo Ren, and Licheng Jiao. Masked angle-aware autoencoder for remote sensing images. In *European Conference on Computer Vision*, pages 260–278. Springer, 2024. 1, 2
- [22] Yang Liu, Chenchen Jing, Hengtao Li, Muzhi Zhu, Hao Chen, Xinlong Wang, and Chunhua Shen. A simple image segmentation framework via in-context examples. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. 3, 5, 6
- [23] Yang Liu, Muzhi Zhu, Hengtao Li, Hao Chen, Xinlong Wang, and Chunhua Shen. Matcher: Segment anything with one shot using all-purpose feature matching. In *ICLR*, 2024. 3
- [24] Utkarsh Mall, Bharath Hariharan, and Kavita Bala. Change-aware sampling and contrastive learning for satellite images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5261–5270, 2023. 1, 2
- [25] Utkarsh Mall, Cheng Perng Phoo, Meilin Kelsey Liu, Carl Vondrick, Bharath Hariharan, and Kavita Bala. Remote sensing vision-language foundation models without annotations via ground remote alignment. *arXiv preprint arXiv:2312.06960*, 2023. 4
- [26] Oscar Manas, Alexandre Lacoste, Xavier Giró-i Nieto, David Vazquez, and Pau Rodriguez. Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9414–9423, 2021. 1, 2
- [27] Matías Mendieta, Boran Han, Xingjian Shi, Yi Zhu, and Chen Chen. Towards geospatial foundation models via continual pretraining. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16806–16816, 2023. 2
- [28] Lingchen Meng, Shiyi Lan, Hengduo Li, Jose M Alvarez, Zuxuan Wu, and Yu-Gang Jiang. Segic: Unleashing the emergent correspondence for in-context segmentation. In *European Conference on Computer Vision*, pages 203–220. Springer, 2024. 3, 5, 6
- [29] Dilxat Muhtar, Xueliang Zhang, Pengfeng Xiao, Zhenshi Li, and Feng Gu. Cmid: A unified self-supervised learning framework for remote sensing image understanding. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–17, 2023. 2
- [30] NASA-IMPACT. Etc-2021-flood-detection. <https://huggingface.co/datasets/blanchon/ETCI-2021-Flood-Detection>, 2021. 3, 5, 6
- [31] Mubashir Noman, Muzammal Naseer, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, and Fahad Shahbaz Khan. Rethinking transformers pre-training for multi-spectral satellite imagery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27811–27819, 2024. 1, 2
- [32] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafranec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research Journal*, pages 1–31, 2024. 3
- [33] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. Pmlr, 2021. 3, 5
- [34] Maryam Rahnemoonfar, Tashnim Chowdhury, Argho Sarkar, Debvrat Varshney, Masoud Yari, and Robin Robertson Murphy. Floodnet: A high resolution aerial imagery dataset for post flood scene understanding. *IEEE Access*, 9:89644–89654, 2021. 6
- [35] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. In *The Thirteenth International Conference on Learning Representations*, 2024. 2, 8
- [36] Tianhe Ren, Yihao Chen, Qing Jiang, Zhaoyang Zeng, Yuda Xiong, Wenlong Liu, Zhengyu Ma, Junyi Shen, Yuan Gao, Xiaoke Jiang, et al. Dino-x: A unified vision model for open-world object detection and understanding. *arXiv preprint arXiv:2411.14347*, 2024. 3
- [37] Zhiyuan Ren, Yiyang Su, and Xiaoming Liu. Chatgpt-powered hierarchical comparisons for image classification. *Advances in neural information processing systems*, 36: 69706–69718, 2023. 4
- [38] Dianmo Sheng, Dongdong Chen, Zhentao Tan, Qiankun Liu, Qi Chu, Tao Gong, Bin Liu, Jing Han, Wenbin Tu, Shengwei Xu, and Nenghai Yu. Unicl-sam: Uncertainty-driven in-context segmentation with part prototype discovery. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 20201–20211, 2025. 3
- [39] Xian Sun, Peijin Wang, Wanxuan Lu, Zicong Zhu, Xiaonan Lu, Qibin He, Junxi Li, Xuee Rong, Zhujun Yang, Hao Chang, et al. Ringmo: A remote sensing foundation model with masked image modeling. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–22, 2022. 1, 2
- [40] Yanpeng Sun, Jiahui Chen, Shan Zhang, Xinyu Zhang, Qiang Chen, Gang Zhang, Errui Ding, Jingdong Wang, and Zechao Li. Vrp-sam: Sam with visual reference prompt. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23565–23574, 2024. 3
- [41] Yao Sun, Yi Wang, and Michael Eineder. Quickquakebuildings: Post-earthquake sar-optical dataset for quick damaged-building detection. *IEEE Geoscience and Remote Sensing Letters*, 21:1–5, 2024. 3, 5
- [42] Wei Suo, Lanqing Lai, Mengyang Sun, Hanwang Zhang, Peng Wang, and Yanning Zhang. Visual prompt selection for in-context learning segmentation. *arXiv preprint arXiv:2407.10233*, 2024. 6
- [43] Benjamin Swan, Joe Pyle, Darrell Roddy, H Lexie Yang, Amy Rose, and Melanie Laverdiere. Orbital-net: A labeled training library for large-scale building feature extraction. *Scientific Data*, 12(1):1650, 2025. 2

- [44] Xin-Yi Tong, Gui-Song Xia, and Xiao Xiang Zhu. Enabling country-scale land cover mapping with meter-resolution satellite imagery. *ISPRS Journal of Photogrammetry and Remote Sensing*, 196:178–196, 2023. 3, 5
- [45] Joshua Veitch-Michaelis, Andrew Cottam, Daniella Schweizer, Eben Broadbent, David Dao, Ce Zhang, Angelica Almeyda Zambrano, and Simeon Max. OAM-TCD: A globally diverse dataset of high-resolution tree cover maps. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2024. 5
- [46] Junjue Wang, Zhuo Zheng, Ailong Ma, Xiaoyan Lu, and Yanfei Zhong. Loveda: A remote sensing land-cover dataset for domain adaptive semantic segmentation, 2021. Preprint at Arxiv <https://arxiv.org/abs/2110.08733>. 3, 5
- [47] Junjue Wang, Weihao Xuan, Heli Qi, Zhihao Liu, Kunyi Liu, Yuhan Wu, Hongruixuan Chen, Jian Song, Junshi Xia, Zhuo Zheng, and Naoto Yokoya. Disasterm3: A remote sensing vision-language dataset for disaster damage assessment and response. In *Proceedings of the Neural Information Processing Systems*, 2025. 1
- [48] Xinlong Wang, Wen Wang, Yue Cao, Chunhua Shen, and Tiejun Huang. Images speak in images: A generalist painter for in-context visual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6830–6839, 2023. 2, 5, 6
- [49] Xinlong Wang, Xiaosong Zhang, Yue Cao, Wen Wang, Chunhua Shen, and Tiejun Huang. Seggpt: Towards segmenting everything in context. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1130–1140, 2023. 2, 5, 6
- [50] Xinye Wanyan, Sachith Seneviratne, Shuchang Shen, and Michael Kirley. Dino-mc: Self-supervised contrastive learning for remote sensing imagery with multi-sized local crops. *arXiv e-prints*, pages arXiv-2303, 2023. 1, 2
- [51] Syed Waqas Zamir, Aditya Arora, Akshita Gupta, Salman Khan, Guolei Sun, Fahad Shahbaz Khan, Fan Zhu, Ling Shao, Gui-Song Xia, and Xiang Bai. isaid: A large-scale dataset for instance segmentation in aerial images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 28–37. IEEE, 2019. 3, 5
- [52] Kang Wu, Yingying Zhang, Lixiang Ru, Bo Dang, Jiangwei Lao, Lei Yu, Junwei Luo, Zifan Zhu, Yue Sun, Jiahao Zhang, Qi Zhu, Jian Wang, Ming Yang, Jingdong Chen, Yongjun Zhang, and Yansheng Li. A semantic-enhanced multi-modal remote sensing foundation model for earth observation. *Nature Machine Intelligence*, 7:1235–1249, 2025. 1, 2, 3, 5, 6
- [53] Aoran Xiao, Weihao Xuan, Junjue Wang, Jiaxing Huang, Dacheng Tao, Shijian Lu, and Naoto Yokoya. Foundation models for remote sensing and earth observation: A survey. *IEEE Geoscience and Remote Sensing Magazine*, 2025. 1
- [54] Zhiyuan Yan, Peijin Wang, Feng Xu, Xian Sun, and Wenhui Diao. Air-pv: A benchmark dataset for photovoltaic panel extraction in optical remote sensing imagery. *Science China. Information Sciences*, 66(4):140307, 2023. 3, 5
- [55] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. 4
- [56] Liang Yao, Fan Liu, Delong Chen, Chuanyi Zhang, Yijun Wang, Ziyun Chen, Wei Xu, Shimin Di, and Yuhui Zheng. Remotesam: Towards segment anything for earth observation. *arXiv preprint arXiv:2505.18022*, 2025. 2
- [57] Renrui Zhang, Zhengkai Jiang, Ziyu Guo, Shilin Yan, Junting Pan, Hao Dong, Yu Qiao, Peng Gao, and Hongsheng Li. Personalize segment anything model with one shot. In *ICLR*, 2024. 3, 5, 6
- [58] Xiaokang Zhang, Weikang Yu, Man-On Pun, and Wenzhong Shi. Cross-domain landslide mapping from large-scale remote sensing images using prototype-guided domain-aware progressive representation learning. *ISPRS Journal of Photogrammetry and Remote Sensing*, 197:1–17, 2023. 6
- [59] Qiqi Zhu, Yanan Zhang, Ziqi Li, Xiaorui Yan, Qingfeng Guan, Yanfei Zhong, Liangpei Zhang, and Deren Li. Oil spill contextual and boundary-supervised detection network based on marine sar images. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–10, 2021. 6
- [60] Qi Zhu, Jiangwei Lao, Deyi Ji, Junwei Luo, Kang Wu, Yingying Zhang, Lixiang Ru, Jian Wang, Jingdong Chen, Ming Yang, et al. Skysense-o: Towards open-world remote sensing interpretation with vision-centric visual-language modeling. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 14733–14744, 2025. 2, 3, 4, 5, 6